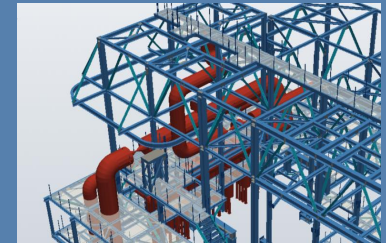
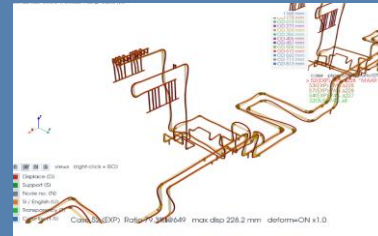
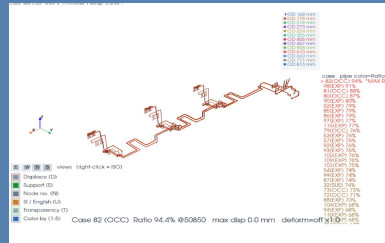
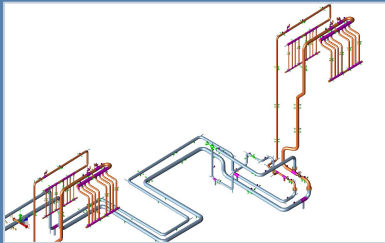


CAESAR II → E3D / SP3D Converter

< Automation of 3D Piping Model from Stress Analysis >

CAESAR II Neutral (.CII) → E3D Macro · AP3D DXF · SP3D IFC



Key Purpose

Why this program

1. View ALL data from the .CII file alone

- No CAESAR II license/program needed — read geometry, supports, sizes directly

2. Convert everything to 3D design formats

- E3D Equipment (MAC) · 3D DXF · SP3D IFC — the whole model, per system

3. Build the Max-Displacement (deformed) 3D model

- Run CLASH CHECK inside Navisworks · AP3D · E3D · SP3D with the deformed shape

Required inputs : CAESAR II Neutral (.CII) + CAESAR II Report + Max Displacement Report

Main Window – Item Guide

The screenshot shows the 'CAESAR II → E3D Equi Macro & Preview Check' window. It features a file list for 'CAESAR II Neutral Files (.CII)' with three files: 'Sample_Line.CII', 'CAESAR REPORT] Stress_Analysis_Report.pdf', and 'MAX-DISP REPORT] Sample_Line_MAX_DIS_REPORT.xlsx'. Below the list are buttons for 'Select CII', 'Remove Selected', 'Clear All', 'Max Displace Report', and 'Print Report'. The 'Output Options' section includes checkboxes for 'Pipework MAC', 'Primitive MAC', 'Deformed-shape MAC', 'Node-number AID tags', and 'Show node labels in 3D preview'. A warning box mentions unit conversion and AID TEXT. The 'Output Folder' section has a text field for the path, a 'Browse...' button, and an 'Open Folder' button. Below this are radio buttons for 'Preview Type' (CII Report, DXF, IFC, Insulation). At the bottom are buttons for 'Build E3D Mac & Converging', 'DXF File Download', 'C File Download', and 'review'. A status bar at the very bottom says 'Ready. Select .CII neutral file(s), then Build or Preview.'

1. Sample_Line.CII

2. Select CII

3. Max Displace Report

4. Print Report

5. Pipework MAC (NEW PIPE/BRANCH routing skeleton)

6. /Project/Output

7. Preview Type: CII Report

8. Build E3D Mac & Converging

9. DXF File Download

10. C File Download

11. review

12. Ready. Select .CII neutral file(s), then Build or Preview.

1. CAESAR II Neutral (.CII) files & directory
2. Select / Remove / Clear .CII files
3. Max Displace Report (Excel/PDF) – deformed source
4. Select CAESAR II Report (PDF)
5. Output Options (Pipework / Primitive / Deformed / AID / labels)
6. Output Folder (Browse / Open Folder)
7. Preview Type (CII / DXF / IFC) + Insulation on/off
8. Build E3D Mac & Converting
9. 3D DXF File Download (Shape + Deform)
10. IFC File Download (Shape + Deform, for SP3D)
11. 3D Preview (CII / DXF / IFC)
12. Log – progress & results

Contents

- **Key Purpose & Main Window**
- **Overview**
- **Input & Options**
- **Output 1 – E3D Macro**
- **Output 2 – 3D Model & Deformed Shape (Clash Check)**
- **Output 3 – 3D DXF**
- **Output 4 – IFC for Smart 3D (SP3D)**
- **Support & Component Shapes**
- **Notes & Limitations**

Overview

View without CAESAR II

- From the .CII file alone — see all geometry / supports / sizes, no CAESAR II needed

One source → many 3D formats

- E3D Equipment (MAC) · 3D DXF · SP3D IFC — whole model, split per system
- Type-specific support shapes; reducer / flange / rigid / expansion distinguished

Deformed model → Clash Check

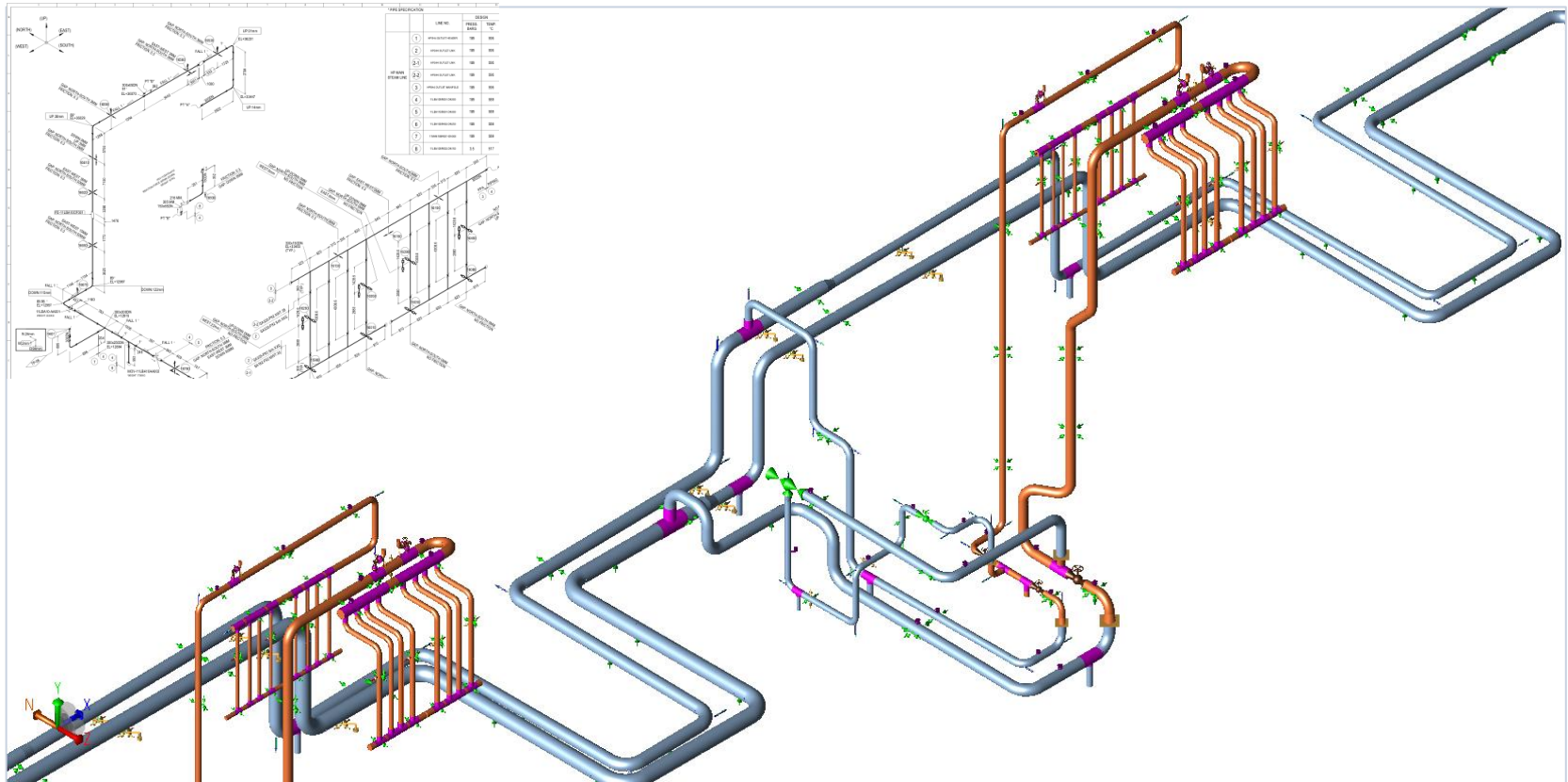
- Max-displacement (deformed) 3D model for clash check in Navisworks / AP3D / E3D / SP3D

Inputs & Portability

- Inputs: .CII + CAESAR II Report + Max Displacement Report — offline, pure-Python GUI

Input & Options

1. CAESAR II Neutral (.CII) File



Select one or more .CII files; the directory is shown next to the title.

Input & Options

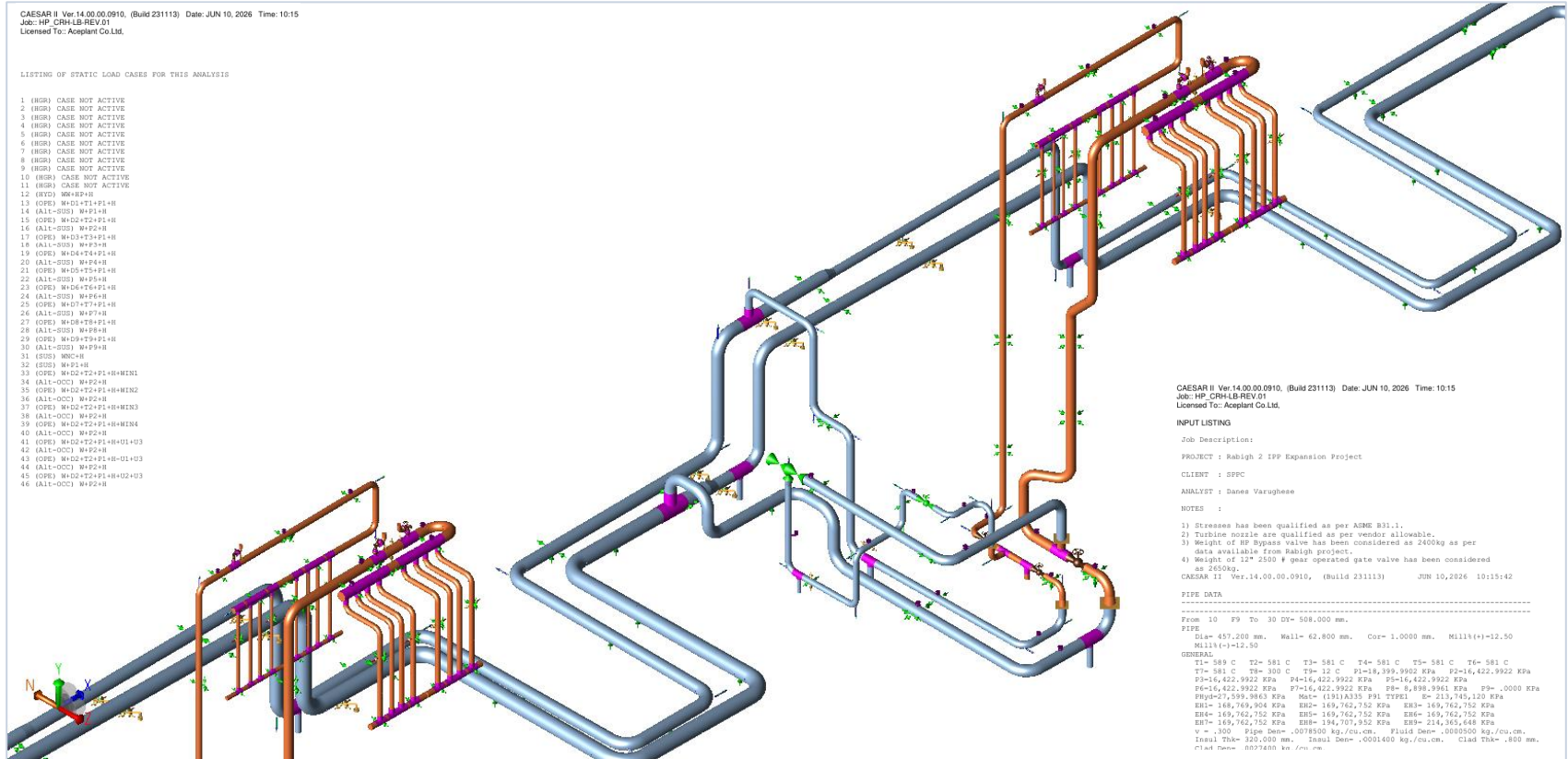
2. Report (Max Displacement / CAESAR Report)



Excel or PDF displacement report drives the deformed-shape output.

Input & Options

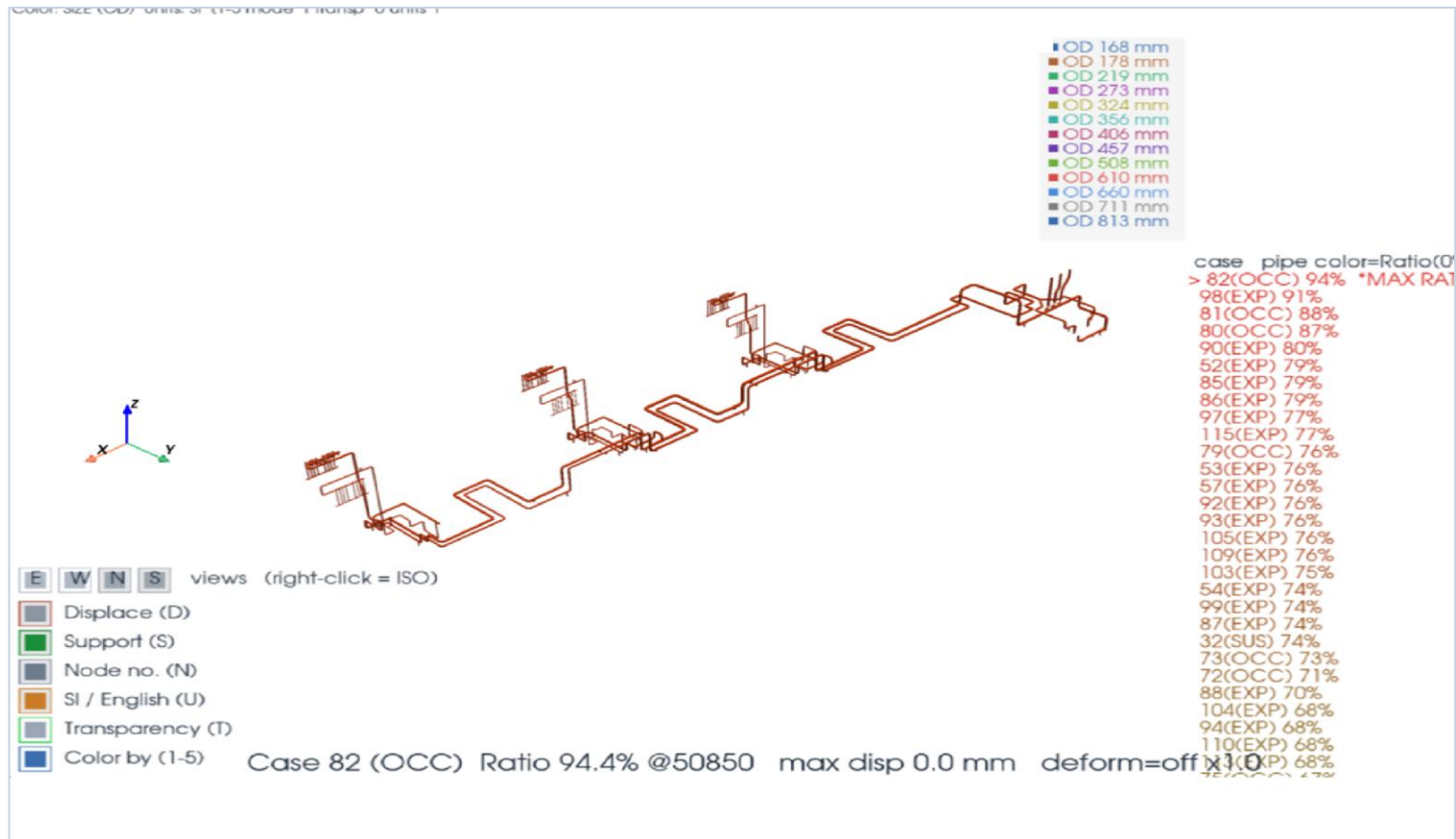
3. Input Options Caesar II Report



Choose Pipework / Primitive / Deformed outputs, node labels, Insulation, and Preview Type (CII / DXF / IFC).

Output – Preview

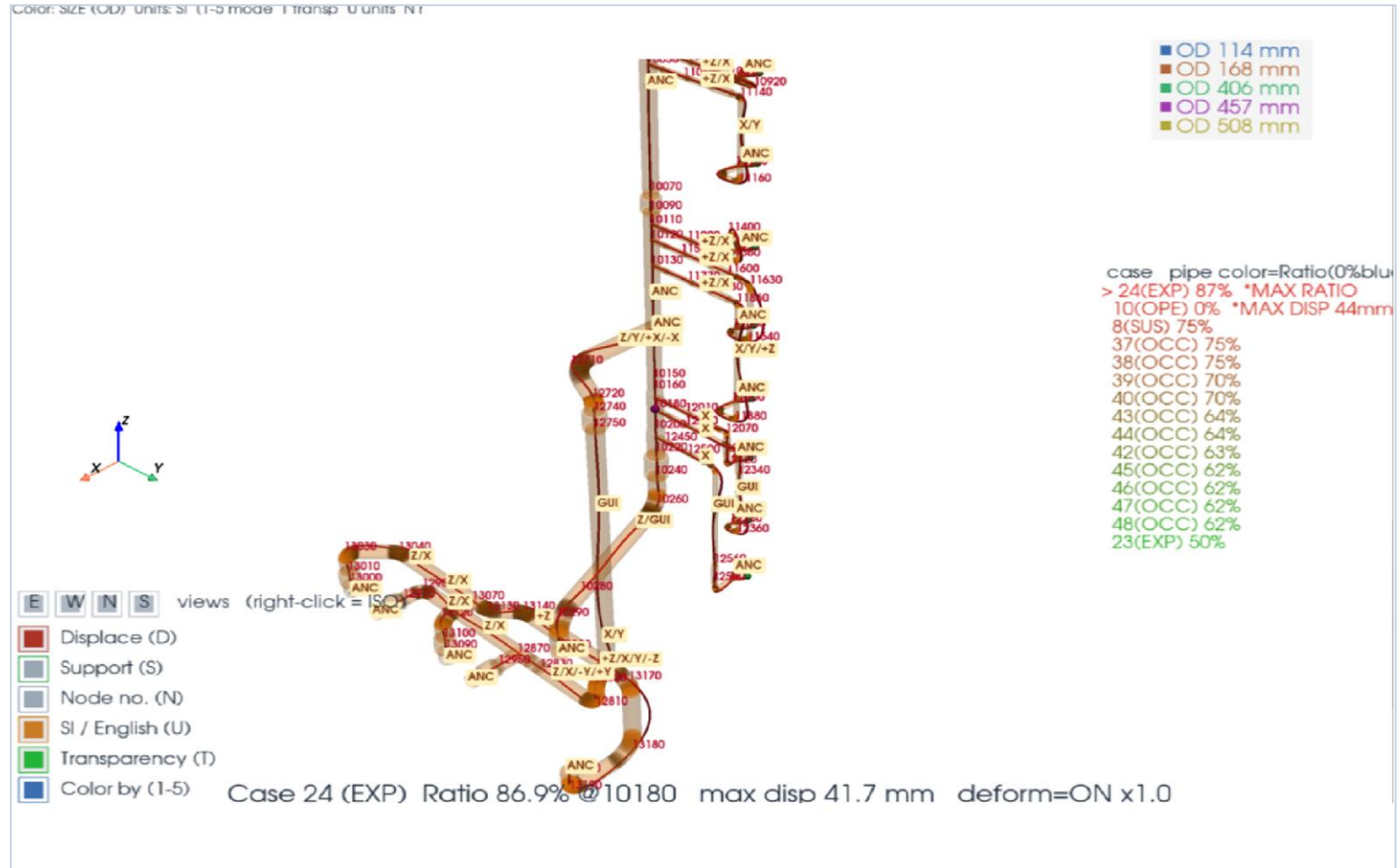
4-1. 3D Report Data Preview



Preview Type: CII Report / DXF / IFC. Faithful shapes, Shape + Deformed together, node find (F).

Output – Preview

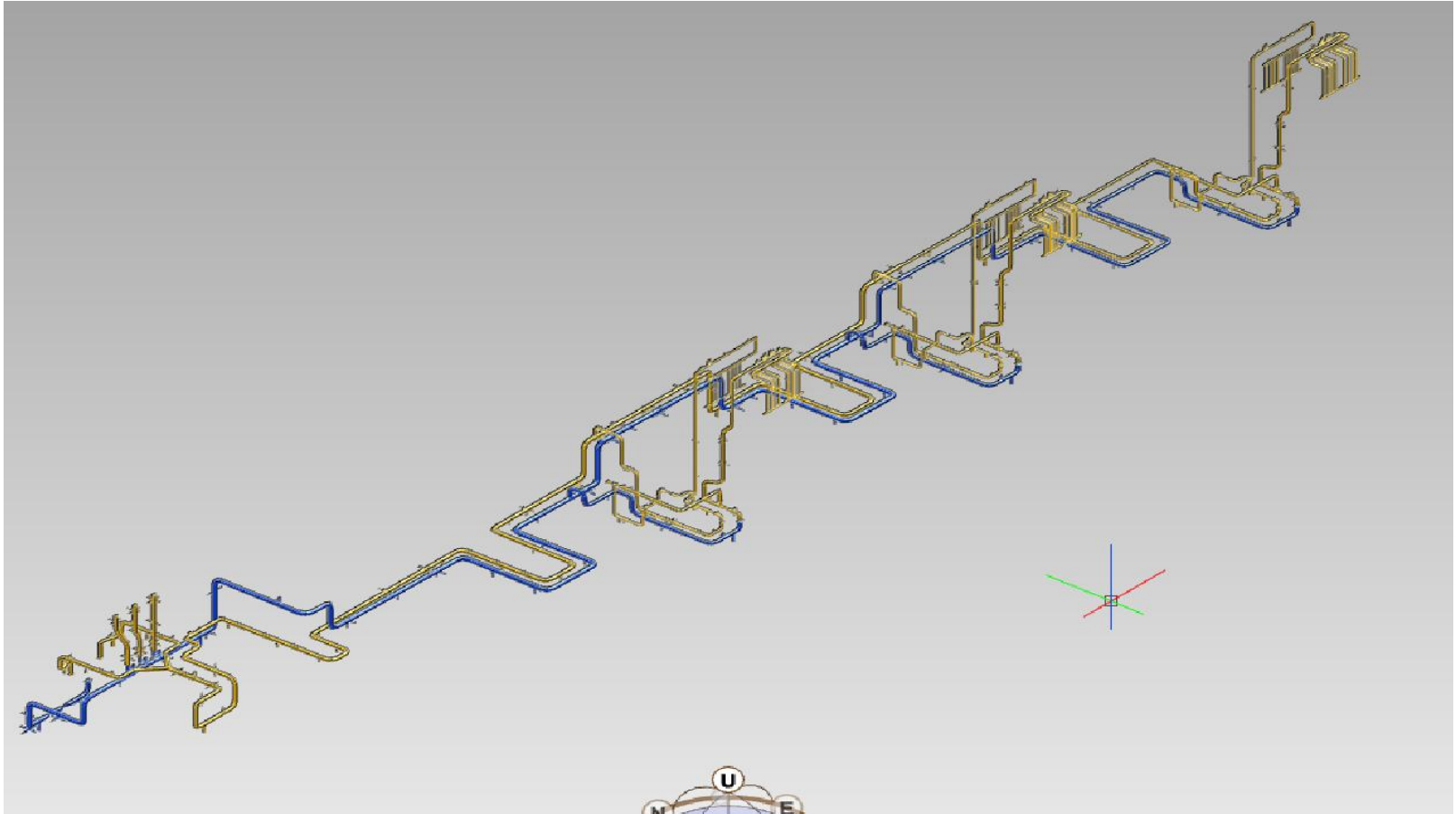
4-2. 3D Report Data Preview



Instant 3D preview: pipes, elbows, supports and node numbers.

Output - E3D Equipment input Macro

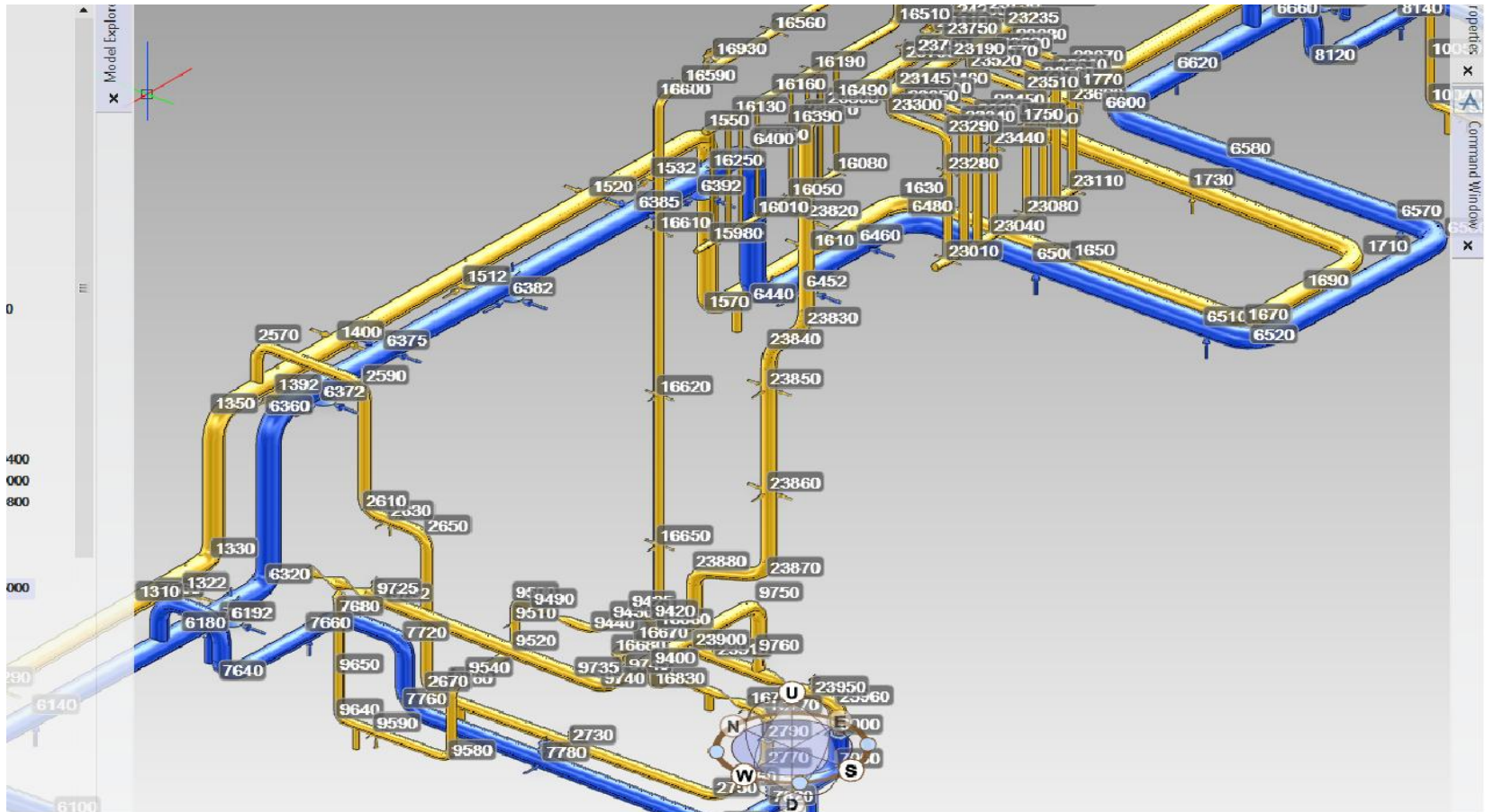
5. E3D Equipment Auto Modeling Input (.MAC)



ZONE > EQUIPMENT (per branch) > SUBE (pipe) + SUBE (each support). Run with \$M in E3D.

Output – E3D Clash with Caesar II Movement Data

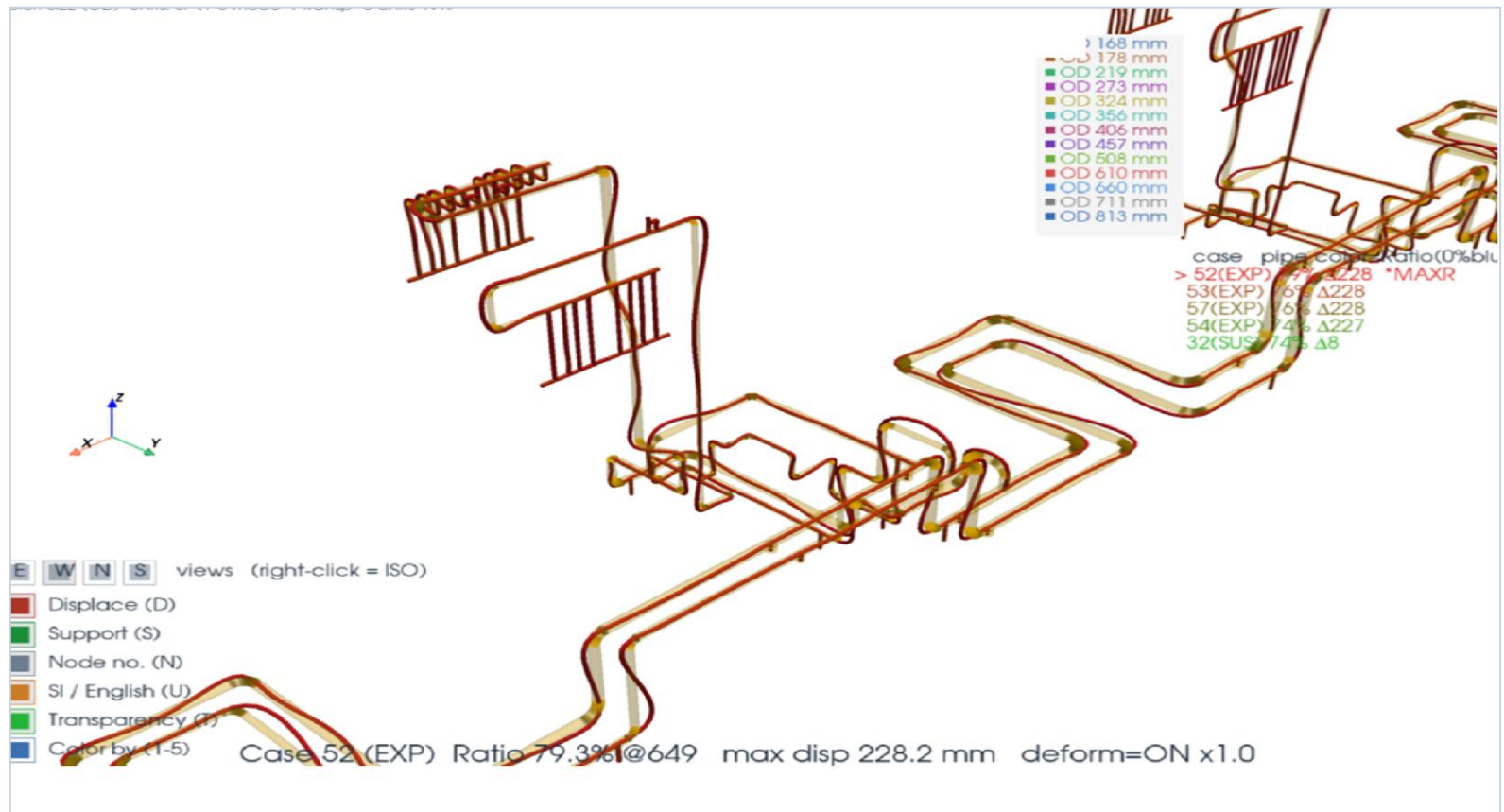
6. Caesar II Shape & Max Disp. Data to E3D Equi Auto Modeling



CYLINDER (pipe) · CTORUS (elbow) · type-specific support solids in E3D.

Output – Movement Data Review

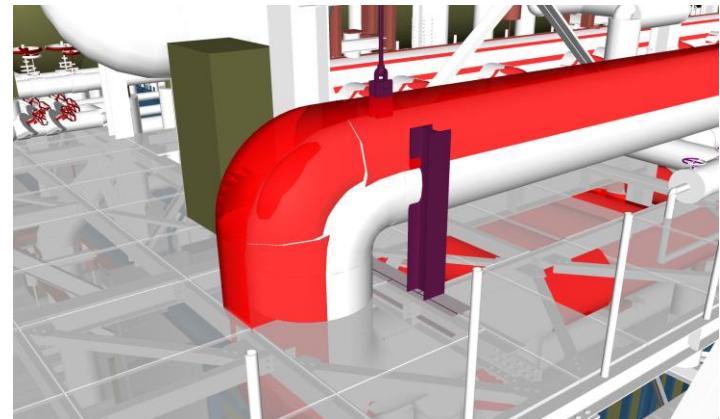
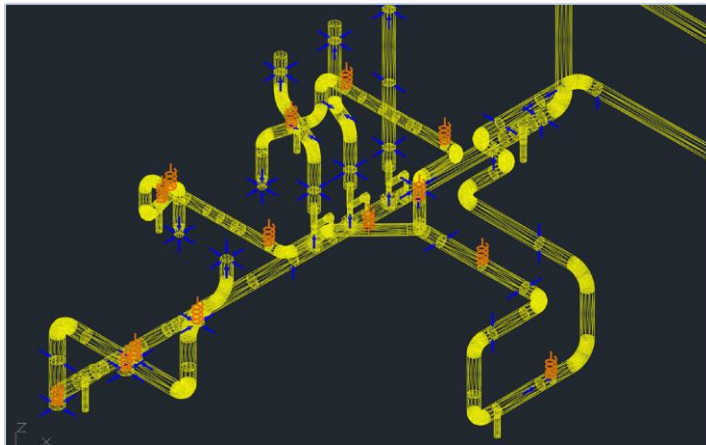
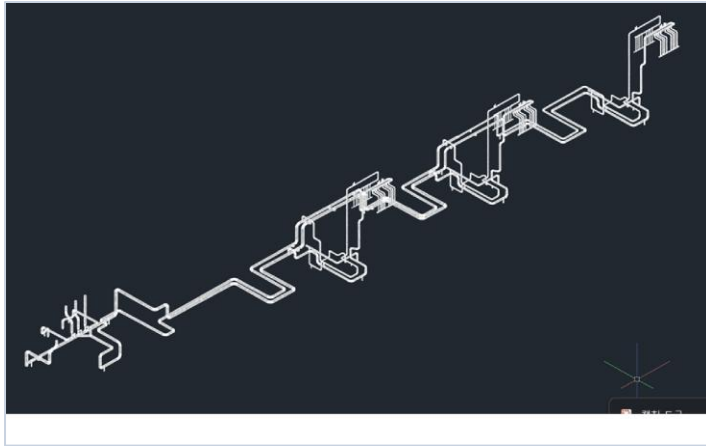
7. Movement Shape Preview → 3D Model Clash Check



Max-displacement deformed model → clash check in Navisworks / AP3D / E3D / SP3D.

Output – 3D DXF for AP3D/NavisWork Review

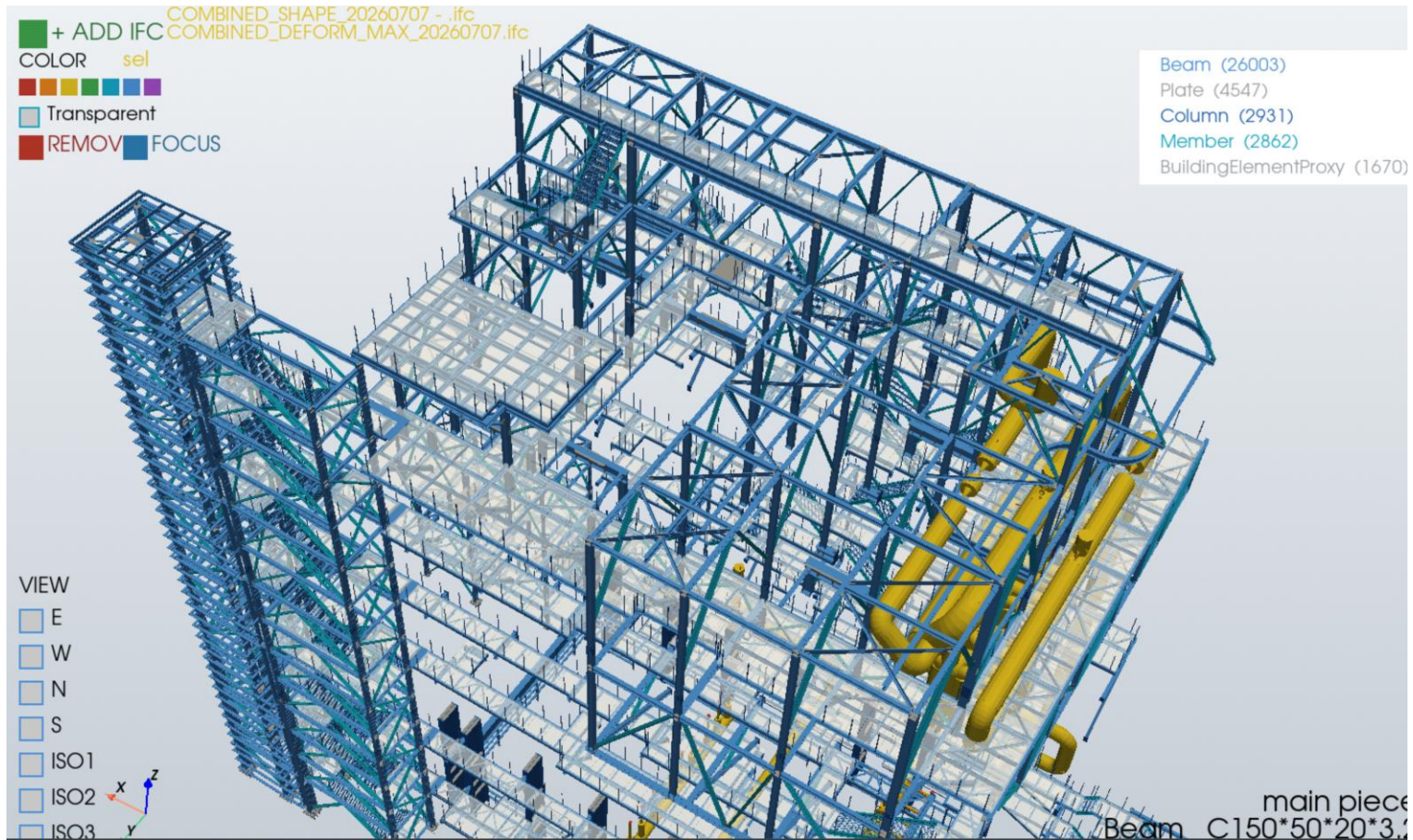
8. 3D DXF (CaesarII Shape & Movement) for Clash Check & Review



3D DXF for CAD / Navisworks / AP3D (no E3D). Layers: PIPE(green) / DEFORM(red) / INSU(gray) / SUPP(blue).

Output – 3D IFC TEKLA

8. 3D IFC (CaesarII Shape & Movement) for Clash Check & Review



IFC solids (IFC2x3) for SP3D via Smart Interop Publisher — verify elbows/reducers before import.

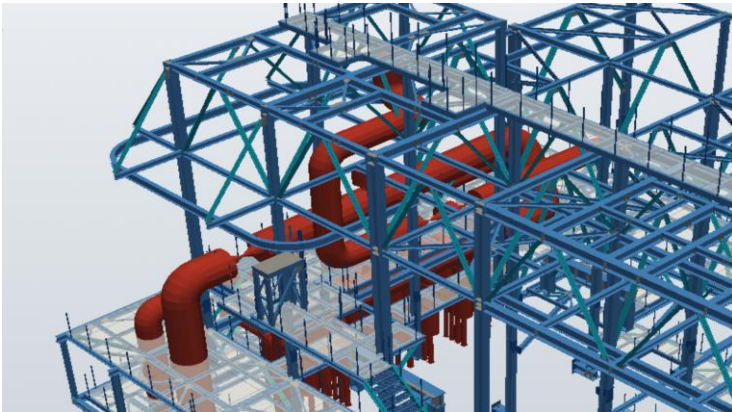
Output – IFC for Smart 3D (SP3D)

For sites that have SP3D (Smart 3D) but no E3D

- Same verified primitive geometry exported as IFC solids (IFC2x3, SI mm)
- Each branch = IfcElementAssembly; each shape = IfcBuildingElementProxy
- Two files: <name>_SHAPE.ifc + <name>_DEFORM_MAX.ifc

Import route

- Load IFC via Smart Interop Publisher, then reference/review in Smart 3D
- Colored by layer: SHAPE green(PIPE) · DEFORM red · INSU gray · SUPP blue
- Geometry validity verified with ifcopenshell



Support & Component Shapes

Item	Shape
Anchor	Base plate (OD x1.1, 10 mm thick)
Rest / Guide	Direction arrow (gap-aware)
Snubber	Shock absorber (rod + body + base)
Spring hanger	Rod + CTORUS coil (3 rings)
Rod / Limit	Slim rod / axial stop plate
Reducer / Flange	Taper cone / disc pair
Rigid (valve etc.)	Bow-tie cones

Notes & Limitations

- Valves cannot be individually identified — the neutral file has no component-type tag (shown as rigid)
- No line numbers / system names — .Cll is node-based (only a few point tags exist)
- E3D MAC AID TEXT / BOX syntax may vary by E3D version — please verify
- IFC is review-grade proxy solids — native piping entities/attributes are future work
- SP3D / SIP import options depend on version (IFC2x3 vs IFC4)

Thank You

CAESAR II → E3D / SP3D · Plant Engineering Total Service System